Soham Manjrekar

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Education

Georgia Institute of Technology Aug. 2024 – Dec. 2025
Master of Science in Electrical Engineering *Thesis Advisor: Dr. Deepak Divan*
- Chair, IEEE Power & Energy Society Student Chapter

University of Illinois Urbana-Champaign Aug. 2020 – May 2024
Bachelor of Science in Electrical Engineering
- Hoeft Technology and Management Minor

Research and Work Experience

Power Electronics Engineer Oct. 2025 – Present
GridTran - Endeavour (Founding Engineer) *Atlanta, GA*
- Designing efficient, scalable, and robust MVSST's for a beachhead market of datacenter power delivery
- Co-led Electrical, Mechanical, and Firmware teams in design, validation and test sprints to rapidly de-risk core technology prototypes for NPI Line

Graduate Research Assistant Aug. 2024 – Oct. 2025
Georgia Institute of Technology – Center for Distributed Energy (CDE) *Atlanta, GA*
- Research Advisor: Dr. Deepak Divan
- Research focused on designing, modeling and prototyping grid-connected inverters which integrate distributed energy resources to be rapidly adopted and scaled for industrial use

Undergraduate Research Assistant Aug. 2023 – May 2024
University of Illinois Urbana-Champaign – Power and Energy Group *Champaign, IL*
- Advisor: Dr. Arijit Banerjee
- Research to experimentally characterize transformer core losses when excited by a signal of multiple frequencies

Electrical Engineering Intern May 2023 – Aug. 2023
Northrop Grumman - Mission Systems *Baltimore, MD*
- Tested and validated Electronic Warfare (EW) systems aboard the EA-18G Growler platform
- Supported design and analysis of the Power Processing Unit (PPU) for electrical systems on the F-35 Lightning II

Electrical Engineering Intern May 2022 – Aug. 2022
Black & Veatch - Power Delivery *Overland Park, KS*
- Physical design of Gas Insulated Substations across the Midwest with delivery to client and performed site surveys thermally validating surge arresters
- Developed proposal for street-side EV charging bays presented to City Council of Kansas City

Electrical Engineering Intern May 2021 – Oct. 2021
American Battery Solutions *Lake Orion, MI*
- Developed an automated system with Python scripts to report requirement testing and validation of battery packs in an ETAS LABCAR HIL
- Built a low-cost CANalyzer alternative using open-source software and economical hardware to transmit CAN signals directly to a mobile app or web server

Projects

Regenerative Clamp Converter - *Thesis Dissertation*
Center for Distributed Energy (CDE), Georgia Institute of Technology *Atlanta, GA*
- Design and analysis of regenerative clamp converter to capture and recover trapped energy in high-power isolated converter topologies

Medium Voltage Inertia-Less Isolated Converter
Center for Distributed Energy (CDE), Georgia Institute of Technology *Atlanta, GA*
- Design, prototyping, and test of a multi-level, medium-voltage, single-stage isolated converter, 800 Vdc to 2.4 kVacrms per slice. Work included topology definition, high-frequency magnetics design, control and protection implementation, hardware and firmware testing and validation.

Integrated Home Energy Fast Charger

Center for Distributed Energy (CDE), Georgia Institute of Technology *Atlanta, GA*

- Design of a 50 kW bidirectional home fast charger, 120 Vac split-phase to 250-400 Vdc, with interleaved PFC, active power decoupling, and isolated DC–DC stage. Supported automatic PV/storage integration, and EV charging using SiC MOSFETs.

Rogowski Coil Analog Current Sensor

Center for Distributed Energy (CDE), Georgia Institute of Technology *Atlanta, GA*

- Design and validation of a wideband, low cost, analog Rogowski coil current sensor for closed loop current control of high frequency power converters.

Single-Stage Isolated DC–AC Energy Router

Center for Distributed Energy (CDE), Georgia Institute of Technology *Atlanta, GA*

- Designed and prototyped a 1 kW single-stage isolated converter based on the inertia-less isolated converter, delivering 240 Vac from a 48 Vdc source with a $\Sigma - \Delta$ modulation scheme.

Machine Learning for Weld Acoustics Monitoring

Caterpillar Inc. and University of Illinois Urbana–Champaign *Peoria, IL*

- Developed a machine learning system to classify weld quality from acoustic signals using MFCC feature pipelines and SVM classifiers; built Python-based GUI feedback, Arduino/LED prototype alert stack, and achieved 75–84% test accuracy on Caterpillar welding lab datasets.

Publications & Patents

[1] Deepak Divan, Joseph Benzaquen, Poornachandra Rao Nallamatti, and **Soham Manjrekar**. *Soft Switching Isolated Inertia-less Modular Multilevel (SSIIMM) Converter*. Provisional Patent Disclosure, GridTran. March, 2026.

[2] Deepak Divan, Joseph Benzaquen, and **Soham Manjrekar**. *Automatic Integration of PV Solar Energy into Multiport Dual Active Bridge Converter-Based EV Chargers*. Provisional Patent Disclosure, Georgia Institute of Technology. June, 2025.

[3] Ruomu Hao, **Soham Manjrekar**, Satish Belkhode, Aniruddh Marellapudi, Joseph Benzaquen, and Deepak Divan. "A Multiport Bidirectional HF-Link Split-phase DC/DC/AC Universal Minimal Converter". In: *IEEE Energy Conversion Congress & Expo (ECCE)*. 2025. DOI: 10.1109/ECCE58356.2025.11259542..

[4] Navami Prabhu, **Soham Manjrekar**, Ritvik Sriram, Aniruddh Marellapudi, Joseph Benzaquen, and Deepak Divan. "Solar Plug – Universal Off-Grid Microconverter for Low-Cost Tier-1-4 Energy Access". In: *IEEE International Decentralized Energy Access Solutions (IDEAS) Conference*. 2025. DOI: 10.1109/IDEAS62193.2025.11157522.

[5] **Soham Manjrekar**, Aniruddh Marellapudi, Satish Belkhode, Joseph Benzaquen, and Deepak Divan. ""AC Cube": A Single-Stage PV/Battery/Grid Energy Router". In: *2025 IEEE Energy Conversion Congress & Exposition Asia (ECCE-Asia)*. Bengaluru, India, 2025, pp. 1–6. DOI: 10.1109/ECCE-Asia63110.2025.11111930.

[6] **Soham Manjrekar**, Poornachandra Rao Nallamatti, Joseph Benzaquen, and Deepak Divan. "A Regenerative Clamp Converter for Energy Capture and Recovery in High-Power Single-Stage Isolated Converters". In: *IEEE Applied Power Electronics Conference & Exposition (APEC)*. March, 2027. Accepted and selected for Dialogue Preview Session (DPS).

Awards

Frank C. Mock Scholarship Dec. 2023

- This scholarship is given in honor of the Frank C. Mock family to be used to help top ECE students

Grainger Power Engineering Award May 2024

- To reward highly qualified and well-motivated undergraduate and graduate students who have chosen to pursue a field of study in electric power engineering

Technical Skills

Modeling & Simulation: MATLAB/Simulink, PLECS, LTspice, PSIM
Hardware Tools: Cadence Allegro/OrCAD, Altium, KiCad, Oscilloscope, Power analyzers, LEM Sensor Platform, TI DSPs (C2000 series), Intel FPGA (DE0-Nano)
Programming: Python (NumPy, scikit-learn, Librosa), C/C++, SystemVerilog, MATLAB, JavaScript, HTML/CSS
Developer Tools: Git, VS Code, Quartus, Code Composer Studio, Eclipse